

NONCOOPERATIVE GAME THEORY

RATIONALITY IN INTERACTION

© Zhenran Guo[†]

[†] SUFE, Shanghai, China

Noncooperative Game Theory : Rationality in Interaction

Zhenran Guo

12th July 2026

Preface

Acknowledgement

This book is written using $\text{\LaTeX}$. The template is based on Agni Datta's "[Preprint Book Manuscript Template](#)", which I have modified to fit my requirements. The famous [ElegantLaTeX](#) template has also inspired me greatly. The source code is available in [my GitHub repository](#) for reference.

Contents

1	Introduction	1
1.1	Game	1
1.2	Common Example of Games	2
I	Strategic Games	3
2	With Perfect Information	5
2.1	Equilibrium	5
2.1.1	Pure Strategy Nash Equilibrium	6
2.1.1.1	Definition	6
2.1.1.2	Examples	6
2.1.1.3	Existence	6
2.2	Mixed Strategy Equilibrium	7
3	With Incomplete Information	9
II	Dynamic Games with Perfect Information	11
III	Dynamic Games with Incomplete Information	13
A	Notation	15
B	Supplementary Scripts	17

Chapter 1

Introduction

In decision theory, we're modeled as acting as if we were maximizing our utility given some exogenous parameters. An example is that producers in competitive markets choose the output that maximizes their profit. In this setting, no individual producer's behavior affects market structure, and expectations about other producers do not influence decision-making.

However, such perspectives of analysis is overly restriction and the real world is not always the case. For example, collusion requires participants to act based on the expectation of other firms' behavior. This is what game theory studies. Game theory is a subject that deals with interaction among decision-makers, focusing on the existence and properties of equilibrium. Equilibrium is a kind of stability, where no participants have incentive to deviate. We'll formalize this concept later on.

1.1 Game

Now we start to consider the formal definition of games. We try to formalize these concepts by analyzing the game of rock-scissors-pape.

Example 1.1.1 (Rock-scissors-pape). In this game, we have 2 players and the consequences can be represented in **Table 1.1**, where 1 stands for win, 0 stands for

draw and -1 stands for lose. This is also called the payoff matrix, which is widely used in game theory. The embedding assumption of this game is that players prefer winning to losing, inducing a preference relation.

		Player 2		
		R	S	P
Player 1	R	0, 0	1, -1	-1, 1
	S	-1, 1	0, 0	1, -1
	P	-1, 1	1, -1	0, 0

Table 1.1: Outcome of the Rock-scissors-pape Game

A game consists of players (along with their preferences) and rules (that maps actions to consequences). Each player i , denoted by A_i , has a set of available actions, called action set $\mathcal{A}_i$. We use a_i to denote a certain action, i.e., $a_i \in \mathcal{A}_i$. As for "rule", it maps the action profile $a = (a_i)_{i \in N}$ to a consequence c , i.e., a mapping g from the Cartesian product of action sets $\mathcal{A} = \prod_{i \in N} \mathcal{A}_i$ to the consequence set C . If we assume the participants are rational, they can surely compare among 2 consequences, inducing a preference $\leq_i^*$ of player i over $C \times C$.

Definition 1.1.1 (Game). A game is a 4-tuple $\langle N, \mathcal{A}, g, \{\leq_i^*\}_{i \in N} \rangle$, where N is the set of player index. If $\leq_i^*$ admit a utility representation, we may also write $\langle N, \mathcal{A}, g, \{u_i\}_{i \in N} \rangle$ instead.

Notice that the consequence and the action is associated with a mapping, we can equivalently define the preference relation over $\mathcal{A} \times \mathcal{A}$. Let a_{-i} denote the action of other players, i.e., $a_{-i} = (a_j)_{j \in N \setminus \{i\}}$, then the preference of player i can be defined as

$$(a_i^{(1)}, a_{-i}^{(1)}) \leq_i (a_i^{(2)}, a_{-i}^{(2)}) : \iff g(a_i^{(1)}, a_{-i}^{(1)}) \leq_i^* g(a_i^{(2)}, a_{-i}^{(2)}).$$

Then a game can be represented by $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$.

Such a game is of the simplest form. In this book, more structure such as information will be added in the following chapters.

1.2 Common Example of Games

Part I

Strategic Games

Chapter 2

With Perfect Information

A strategic game is a scenario where players make decisions independently and simultaneously. For perfect information, consider an experimental setting, every subject, anonymously sitting behind a computer, is shown a payoff matrix and asked to choose their strategy. They won't be able to see the outcome until everyone makes their choice. To make a correct decision, they should care not only about their own choice, but also make decisions under other subjects' possible action, i.e., form a belief. To be more specific, we need to consider two things: how should one interact given opponents' action? And, what are opponents' possible actions? We will formalize this idea soon after we give the definition of a strategic game.

Definition 2.0.1 (Strategic Game). A strategic game is a 3-tuple $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$. Everything is defined in the introduction part of the book. If every $\mathcal{A}_i$ is finite, we say the game is finite.

2.1 Equilibrium

In this section, we make clear step one of decision choice, i.e., given your opponents' choice, how you should behave.

2.1.1 Pure Strategy Nash Equilibrium

2.1.1.1 Definition

Since we assume players only care about their own payoff, this is equivalent to solving:

$$\max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}), \text{ where } a_{-i} \in \prod_{j \neq i} \mathcal{A}_j \text{ is a given parameter.}$$

We denote its solution correspondence $BR_i(a_{-i})$, called the best-response correspondence.

We refer to the steady state where every subject is playing a best-response equilibrium. It's important to notice that in the context of strategic game, players may play multiple rounds of game, but the rounds are independent, meaning there's no strategic link between rounds.

Definition 2.1.1 (Pure Strategy Nash Equilibrium(PSNE)). Denote $BR(a) := \prod_{i \in I} BR_i(a_{-i})$. We say a^* is a PSNE if $a^* \in BR(a^*)$. Or equivalently, a^* is a PSNE if

$$\forall i \in N : (a_i, a_{-i}^*) \leq (a_i^*, a_{-i}^*), \forall a_i \in \mathcal{A}_i.$$

This leads to an important way of finding the equilibrium:

- STEP I: find the best-response correspondence of each player.
- STEP II: find the action profile satisfying $a_i^* \in BR_i(a_{-i}^*)$.

2.1.1.2 Examples

BoS; Coordination Game; PD; H-D; Pennis(strictly competitive game)

2.1.1.3 Existence

As we've already seen, not every strategic game has a PSNE. In this subsection, we provide some conditions that guarantee the existence. By the definition of PSNE, it's natural to use the correspondence version of fixed point theorem. Here we use Kakutani's fixed point theorem, which says that as long as the correspondence is nonempty, closed and convex, there exists a fixed point.

To suffice this condition, we assume the followings:

- Action sets are nonempty compact convex subset of $\mathbb{R}^n$;
- The preference relation is continuous and quasi-concave on $\mathcal{A}_i$.

Then by Kakutani's fixed point theorem, there exists PSNE.

Theorem 2.1.1. *For a strategic game $\langle N, \mathcal{A}, \{\preceq_i\}_{i \in N} \rangle$, suppose for all $i \in N$, $\mathcal{A}_i$ is nonempty, compact and convex. Further, we suppose $\preceq_i$ is continuous and quasi-concave on $\mathcal{A}_i$, then there exists PSNE.*

2.2 Mixed Strategy Equilibrium

For many games, sticking with one strategy is somehow stupid. It's like keep playing scissor in the rock-scissor-pape game, which is clearly unwise. So, players may deliberately randomize the strategy played.

Other motivating example concerning the introduction includes the matching pennis, which is similar to the rock-scissor-pape game, and the relationship between auditors and taxpayers, see [Osborne and Rubinstein, 1994](#), (p. 37).

Chapter 3

With Incomplete Information

Part II

Dynamic Games with Perfect Information

Part III

Dynamic Games with Incomplete Information

Appendix A

Notation

Appendix B

Supplementary Scripts

Bibliography

Fudenberg, D. and Tirole, J. (1991). *Game Theory*. The MIT Press. MIT Press, London, England.

Osborne, M. J. and Rubinstein, A. (1994). *A Course in Game Theory*. MIT Press, London, England.